

Deterministic Optimization

Algebraic Aspects of Linear Optimization

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Standard Form LP

Algebraic Aspects of Linear Optimization

Learning Objective for this lesson

- Identify a standard form of writing an LP

Standard Form LP

- A **Standard Form** linear program is written as
 - $\min c^\top x$
 - s.t. $Ax = b$  Data dependent
 - $x \geq 0$  Data independent and has simple structure
- Equality constraints
- Nonnegative constraints on all variables

Convert to Standard Form LP

- Every LP can be written equivalently in standard form:

- $\max c^\top x$
- $a_i^\top x \geq b_i$
- $a_i^\top x \leq b_i$
- $x_j \leq 0$
- x_j free

- $-\min -c^\top x$
- $a_i^\top x - s_i = b_i, s_i \geq 0$
- $a_i^\top x + s_i = b_i, s_i \geq 0$
- $x_j' \geq 0$, replace x_j with $-x_j'$
- Replace $x_j = x_j^+ - x_j^-$, $x_j^+ \geq 0, x_j^- \geq 0$

Example

- Original LP:

- $\max x_1 - x_2$
- S.t. $x_1 + x_2 \leq 2$
- $2x_1 - x_2 \geq 3$
- $x_1 \leq 0, x_2 \text{ free}$

- Intermediate step:

- $-\min -x_1 + x_2$
- S.t. $x_1 + x_2 + s_1 = 2$
- $2x_1 - x_2 - s_2 = 3$
- $x_1 \leq 0, x_2 \text{ free}$

- Equivalent standard form LP:

- $-\min x_1 + x_2 - x_3$
- S.t. $-x_1 + x_2 - x_3 + s_1 = 2$
- $-2x_1 - x_2 + x_3 - s_2 = 3$
- $x_1, x_2, x_3, s_1, s_2 \geq 0$

Summary

- We learned:
 - What is standard form LP
 - How to transform any linear program into a standard form LP
- Advantage of standard form LP:
 - Complicating constraints are all equality
 - Only inequality constraints are simple nonnegativity constraints, which do not depend on data